EXPERIMENT NO 8

8.1.1 Multiplication of two square matrices

Algorithm

Step 1: Start the program.

Step 2: Input the dimension n of the square matrices.

Step 3: Input the elements of the first matrix A of size n × n.

Step 4: Input the elements of the second matrix B of size n × n.

Step 5: Initialize i = 0.

Step 6: Check if i < n.

- If **No**, go to **Step 18**.
- If **Yes**, go to **Step 7**.

Step 7: Initialize j = 0.

Step 8: Check if j < n.

- If **No**, go to **Step 16**.
- If **Yes**, go to **Step 9**.

Step 9: Set C[i][j] = 0.

Step 10: Initialize k = 0.

Step 11: Check if k < n.

- If **No**, go to **Step 14**.
- If **Yes**, go to **Step 12**.

Step 12: Compute
C[i][j] = C[i][j] + A[i][k] × B[k][j].

Step 13: Increment k = k + 1 and go back to **Step 11**.

Step 14: Increment j = j + 1.

Step 15: Go back to **Step 8**.

Step 16: Increment i = i + 1.

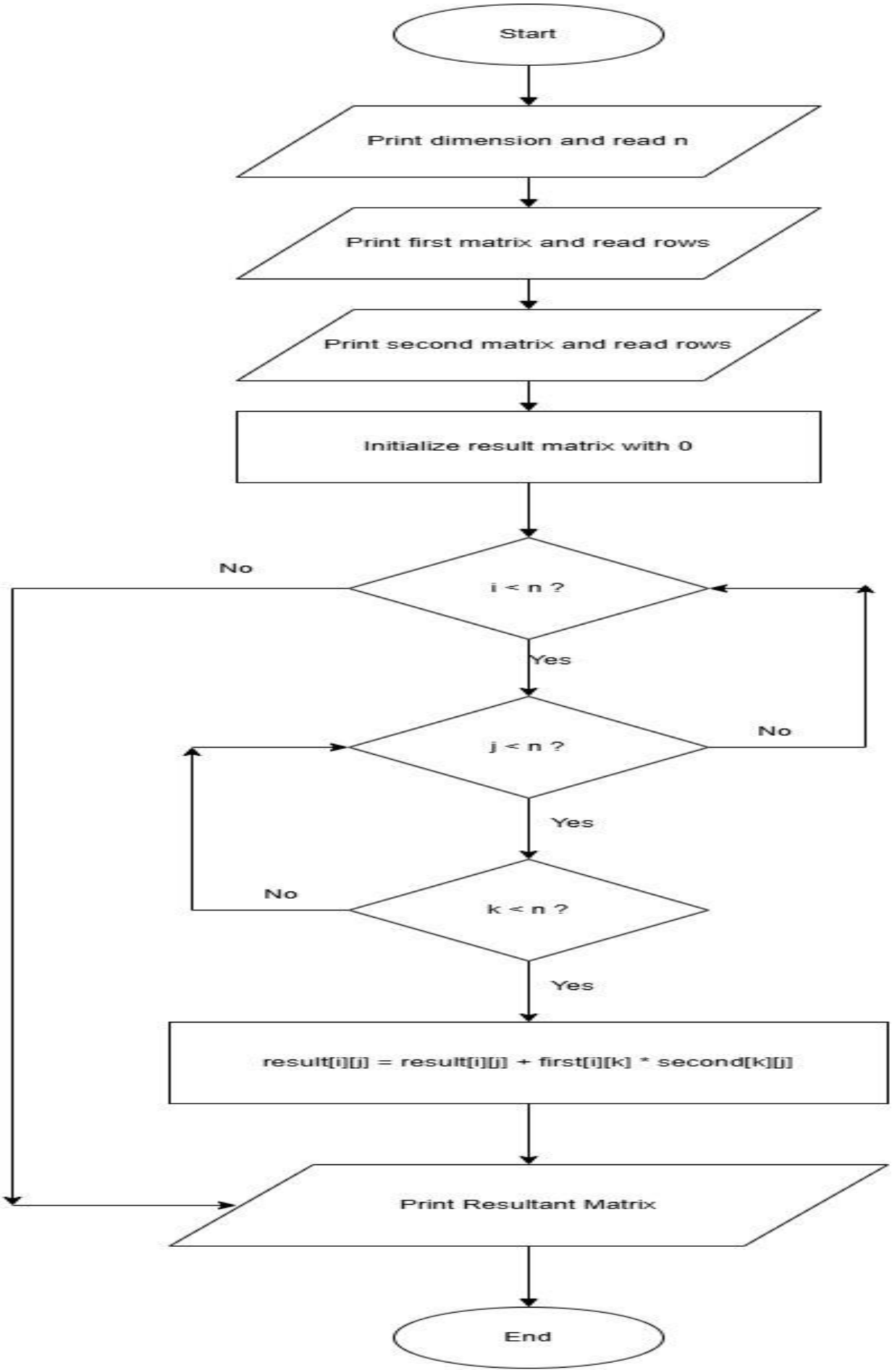
Step 17: Go back to **Step 6**.

Step 18: Print the resultant matrix C.

Step 19: End the program.

EXPERIMENT NO 8

Flowchart



EXPERIMENT NO 8

PYTHON CODE

```
n = int(input("dimension: "))
print("first matrix:") A = [] for i in
range(n):     row = list(map(int,
input().split()))     A.append(row)
print("second matrix:") B = [] for i in
range(n):  row = list(map(int,
input().split()))  B.append(row)
result = [] for i in
range(n):
result.append([0] * n)
for i in range(n):     for j
in range(n):        for k in
range(n):
          result[i][j] += A[i][k] * B[k][j]


print("Resultant Matrix:")
for row in result:
print(*row)
```

EXCECUTION

EXPERIMENT NO 8

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8.1.1. Multiplication of Two Square Matrices

01:57

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices $(n \times n)$.
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases +

Explorer

Multiplica...

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```python
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    row = list(map(int, input().split()))
    A.append(row)

# Read second matrix
print("second matrix:")
B = []
for i in range(n):
    row = list(map(int, input().split()))
    B.append(row)

# Initialize result matrix with zeros
result = [[0 for _ in range(n)] for _ in range(n)]

# Multiply matrices
for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j] += A[i][k] * B[k][j]

# Print result
print("Resultant Matrix:")
for row in result:
    print(*row)
```

Debugger

Terminal Test cases

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